# Statement of Teaching Interests and Philosophy

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## Teaching interests and course alignment

My primary interest is in communication systems, signal and image processing, and machine learning/artificial intelligence. My PhD in Electrical and Computer Engineering focused on AI/ML for wireless networks; I previously taught undergraduate Computer Programming, Communication Systems and Wireless Communication, and I currently work on telecommunications machine-learning engineering at AT&T. I would prioritize the following Rowan University courses, subject to current departmental outlines and section requirements:

Strongest alignment — first-choice assignments

• ECE 09433 Electrical Communication Systems — previously taught Communication Systems; strong preparation in modulation, link performance, digital communications and simulation.

• ECE 09432 Wireless Communications — previously taught Wireless Communication; doctoral research and industrial wireless/network experience.

• ECE 09341 Signals and Systems — foundation in Fourier/Laplace analysis, transforms, linear systems, communication signals and computational examples.

• ECE 09351 Digital Signal Processing — DSP coursework and applied signal-analysis research, with MATLAB/Python demonstrations and discrete-time filtering exercises.

• ECE 09452 Introduction to Digital Image Processing — deep-learning, computer-vision and signal-processing preparation; practical image enhancement, feature extraction and recognition assignments.

• ECE 09455 Machine Learning and ECE 09454 Introduction to Modern Neural Networks — extensive graduate research and current industry application of ML, neural networks, evaluation, and reproducible Python/PyTorch projects.

Additional courses with refresher or laboratory preparation

• ECE 09241 Introduction to Digital Systems — foundational digital logic and computation background; willing to review the current hardware, implementation platforms, and laboratory expectations.

• ECE 09321 Systems and Control I — formal undergraduate/graduate coursework in control systems; requires a targeted refresher in classical analysis, MATLAB demonstrations, and Rowan-specific lab requirements.

• RF/microwave systems, antennas and electromagnetics, if needed — earlier bachelor's/master's coursework and extensive wireless-network research provide a foundation, but I would update specialized RF design, instrumentation and lab techniques before teaching.

• Selected applied engineering projects, engineering clinics, and interdisciplinary AI-for-ECE topics, depending on student needs and available supervision.

This distinction is intentional: prior classroom instruction, current research/industry expertise and willingness to prepare for adjacent subjects are different forms of qualification. I would coordinate course selection and preparation with the department.

## Teaching philosophy: analytical foundations through implementation

I believe engineering students understand complex systems best when mathematical analysis, working examples and design decisions reinforce one another. My classes would progress from clear learning objectives and concise derivations to software-based experiments and interpretation of results. In communications, students might vary signal-to-noise ratio and compare bit-error-rate predictions with simulations; in DSP, they might design and test filters on a measured signal; in image processing, they might compare enhancement methods; and in ML, they might reproduce a baseline and diagnose errors. Each activity would culminate in an explanation of why the method works, its assumptions and limitations, not just a successful plot or code run.

The approach is grounded in my prior teaching at the University of Lahore and communications laboratory support at UET Taxila. Students arrive with varied backgrounds in mathematics, programming and circuit theory. I would start with diagnostic questions and scaffolded tasks, provide worked examples and accessible office hours, and progressively remove scaffolds as learners develop confidence. Rubrics would assess conceptual understanding, implementation correctness, experimental reasoning, teamwork and technical communication.

## Experiential learning, assessment and inclusion

Rowan's engineering-clinic culture and lab/project integration match my interest in hands-on, inquiry-driven teaching. I would use small, achievable teams and clearly defined individual contributions; Git-based version control, reproducible notebooks or MATLAB scripts, structured lab reports, and short design reviews where appropriate. Example projects would rely on available department resources and approved safety practices rather than assuming access to specific equipment.

I would combine low-stakes checks for understanding with laboratory milestones, programming assignments, midterm evaluations and a culminating application. Frequent feedback would help students correct misconceptions early; multiple examples and accessible instructions would support students with different academic preparation. Students should be able to articulate tradeoffs, evaluate results honestly, and identify when a method or model is unreliable.

## Industry-connected mentoring

At AT&T, my work involves AI-supported telecommunications service assurance, network diagnostics and root-cause classification, including a workflow that improved automated classification performance from approximately 20% to 80%. My doctoral research involved wireless simulation, data-efficient ML, and network reliability. Prior roles at AT&T Labs, Futurewei and Impac reinforced the importance of data quality, experimental design, collaboration and reproducibility. I would translate those experiences into de-identified or synthetic student datasets and assignments, never proprietary or sensitive operational records.

My goal is for students to leave each course able to derive key ideas, build and test working implementations, interpret engineering evidence, and communicate reliable conclusions. I would be glad to develop the specific assignments and project scope collaboratively with Rowan's ECE faculty.